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Resilient Consensus Control for Linear Multi-agent System Against the False Data Injection Attacks

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Abstract: This paper is concerned with the resilient consensus control problem of the linear multi-agent system under the false data injection attacks, where each agent may receive the injected false data by attackers from its neighbours through communication links. Due to the existence of the false data, the desired consensus of multi-agent can not be reached by the general distributed consensus algorithms. By constructing an extended state observer to estimate the state and the injected false data towards each agent, a distributed resilient consensus control algorithm is designed to offset the negative effect of the attacks. Firstly, a sufficient condition is derived for the undirected multi-agent system under a bounded and decaying false data injection. Secondly, a sufficient and necessary condition is provided for the directed multi-agent system under the false data injection with decaying rate. Finally, the simulation results illustrate the effectiveness of the proposed consensus algorithms against the false data attacks.

Keywords: Distributed control, extended state observer, false data injection attacks, multi-agent system, resilient consensus.

1. INTRODUCTION

Over the past decades, the distributed control of multi-agent systems (MASs) has captured much attention due to its vast applications in a variety of areas such as consensus control, flocking control, formation control, containment control, and distributed sensor estimation network [1-3], among which consensus control is a fundamental distributed algorithm reaching the agreement of agents' states in MASs [4-8]. While during the procedures of information transmission or the information processing of the communication network of MASs, the collected data without security protection are more vulnerable to malicious cyber attacks. As a result, the performance of MASs may be dramatically destroyed and the desired control objective can not be achieved anymore without the corresponding detection and defense strategies.

Generally, there are mainly two types of cyber attacks, including denial of service (DoS) attacks and deception attacks [9,10]. DoS attacks mainly focus on cutting off the information flow within the communication networks, while deception attacks often tamper the reliability of the real-time data. The false data injection attacks (FDIAs) can be categorized as one kind of deception attacks where the false data is injected into the communication links or the communication nodes such that the agents receive the wrong real-time data instead of the original actual data of agents and then make the wrong group decisions. FDIA is a more insidious attack that destroys the integrity and accuracy of data compared with DoS attack. As for DoS attacks, many researchers have made great efforts to study them from different perspectives recent years. For example, the DoS attacks' detection and defense systems have been investigated in [11,12]; the problems of distributed/remote state estimation under DoS attacks were investigated in [13,14].

On the other hand, lots of efforts have been devoted to the investigation on how the attacker should optimize its attack schedule in order to maximize the effect on the system performance [15,16]. Regarding the consensus problems on the DoS attacks, some researchers characterized frequency and duration of the DoS attacks such that the stability of the closed-loop system can be preserved [17,18]. The secure control problems were solved in [19] by resorting the stochastic analysis methods. The consensus problems against DoS attacks were studied from the perspective of designing the distributed controller to cope with the attacks [20-22].

Simultaneously, FDIAs have also received considerable attention, especially in various application scenarios of smart-grid [23-28], which plays a huge role in modern society, and its disruption may lead to potentially seri-

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ous damages in terms of economic, environmental and social costs. As a result, it has become a preferred target for an increasing number of potential attackers. A major feature of the smart grid is its wide geographic distribution and complex interdependence among different components. In addition, the interruption of connections with modern power grids can leave the internet vulnerable to different kinds of cyber attacks, so there exist many challenges to protect the smart grid from malicious attacks.

The FDIAs in physical spaces were introduced in the fields of electrical power grid in [23]. Several detection algorithms for FDIAs have been developed in the last few years which were either model-based or data-driven algorithms [24-26]. Musleh *et al*. [26] provided an intensive summary of these algorithms by categorizing them and elaborating on the pros and cons of each category. On the other hand, Hao *et al*. [27] and Bae *et al*. [28] proposed an adaptive markov defending strategy and a cost-effective placement method of phasor measurement units respectively to protect the smart grid system against FDIAs.

Although there have been some results on detection and defence strategies for smart grid under FDIAs, the relative research results about consensus control of MASs under FDIAs are still very few. The work presented in [9,29-32] considered the consensus of MASs under randomly occurring FDIAs. The authors in [33-35] discussed the consensus problem in Markov jump systems with FDIAs. According to the existing literatures, the authors in [9,29] investigated the consensus maintenance problems of MASs in the presence of attacks by designing a distributed impulsive controller to achieve the bounded synchronization. An adaptive attack compensator was proposed in [36-38] for MASs, which could ensure the reliability of the system when the false data was injected.

Among the consensus control problems mentioned above, resilient control is widely used in countering uncertainties, constraints and adversarial behaviors. The ability to maintain state awareness in the face of unexpected errors and threats is a defining feature of a resilient control system. In MASs, the parameters of the system usually undergo random variations due to the complex environments, failures, and repairs of components. Some researchers studied the system uncertainties by a resilient control system [40-43]. On the other hand, physical constraints occur frequently, such as actuator saturation and network bandwidth constraints. [39,44]. Thus, it is practically significant to take the corresponding constraints into account during the in-depth investigation of the resilient control problem for MASs. The resilient control design was widely applied when the MASs were subjected to the cyber attacks, faults and disturbances [38,45-47].

In the actual MASs, the FDIAs can occur in the transmission channels of any layer. Most of the published work consider the FDIA on the controller, actuator, sensor or observer, while few results pay close attention to the scenario that the false data is injected to the communication links from neighbouring agents in network layer. Inspired by the above discussions, this paper is concerned with the resilient consensus control problems of linear MASs against the FDIAs from the neighbours' communication links, which has not been considered in the previous work.

The main contributions of this paper are provided as follows: 1) Different from the previous published results, it is assumed that the agents receive the false data from their neighbours, and then an extended state observer is introduced to estimate the state and false data of the current agent. Based on the estimated values, a distributed resilient consensus controller is designed to eliminate the impact of the attack. 2) Two types of false data are discussed. For the undirected MASs, the resilient consensus strategy is built when the false data is injected to the communication links with an attenuated tendency; while for the directed MASs, the consensus strategy is built when the false data is injected to the neighbours' agents with an attenuated rate.

Notations: Let $\mathcal{G}(\mathcal{V},\mathcal{E},\mathcal{A})$ be a graph, where $\mathcal{V} = \{v_1, v_2, \cdots, v_N\}$ is the set of nodes, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges, and $\mathcal{A} = \{a_{ij}\}$ is a weighted adjacency matrix. An edge of $\mathcal{G}(\mathcal{V},\mathcal{E},\mathcal{A})$ is denoted by $e_{ij} = (v_i, v_j)$. The adjacency elements a_{ij} are nonnegative, where $a_{ij} > 0$ if and only if $e_{ij} \in \mathcal{E}$, else $a_{ij} = 0$. For the ith node, the set of neighbours is defined by $N_i = \{v_j \in \mathcal{V}(v_i, v_j) \in \mathcal{E}\}$. The $\mathcal{G}(\mathcal{V},\mathcal{E},\mathcal{A})$ is called connected if there exists a path between any two nodes. The Laplacian matrix $L = \{l_{ij}\}_{N\times N}$ is defined as $l_{ij} = -a_{ij}, i \neq j$, $l_{ii} = \sum_{j=1, j\neq i}^{N} a_{ij}$.

Compared with the undirected graphs, the digraph is more common. Let $\mathcal{G}(\mathcal{V},\mathcal{E},\mathcal{A})$ be a digraph, and the in-degree and out-degree of the node v_i are defined as

$$deg_{in}(v_i) = \sum_{j=1}^{N} a_{ji}, \quad deg_{out}(v_i) = \sum_{j=1}^{N} a_{ij}.$$

The node v_i is balanced if and only if its in-degree and out-degree are equal, ie. $deg_{in}(v_i) = deg_{out}(v_i)$. The digraph is called balanced if and only if all of its nodes are balanced. A node is said to be the root node if it can reach any other nodes through a directed path. A digraph is called to have a directed spinning tree if there exists a root node such that all other nodes can be linked to the root node via a directed path. For a certain matrix, it is Hurwitz only if all of its eigenvalues exist negative real parts. A matrix A is *irreducible* if there doesn't exist a permutation matrix P such that PAP^T is block triangular.

An outline of this paper is as follows: In Section 2, problem formulation and some preliminaries are provided. Section 3 focuses on resilient consensus control problems of linear MASs under FDIAs in undirected or digraph MASs, respectively. Section 4 presents the simulation results and Section 5 draws the conclusions.

2. PROBLEM FORMULATION AND PRELIMINARIES

2.1. Problem formulation

Consider the general linear MAS composed of N agents and ith agent's dynamics is described by

$$\begin{cases} \dot{x}_i(t) = Ax_i(t) + Bu_i(t), \\ y_i(t) = Cx_i(t), \end{cases} \tag{1}$$

where $x_i(t) \in R^n$ and $y_i(t) \in R^m$ are respectively the state variable and output variable, and $u_i(t)$ is the control input of the agent. The coefficient matrices are $A \in R^{n\times n}$, $B \in R^{n\times q}$, $C \in R^{m\times n}$. It is further assumed that (A,B,C) is stabilizable and detectable.

The general distributed consensus protocol $u_i(t) = u_i^{(1)}(t)$ based on the state feedback is

$$u_i^{(1)}(t) = \sum_{j=1}^{N} a_{ij}K\big(x_i(t) - x_j(t)\big), \tag{2}$$

where a_{ij} is the element of the weighted adjacent matrix $\mathcal{A}$ of the communication topology among the agents, and K is the feedback gain matrix.

It has been shown that the system (1) under the protocol (2) is consensusable if the undirected communication topology of MASs is connected and the directed communication topology of MASs has a directed spanning tree [50].

Suppose the MASs are subjected to the FDIAs on the network, and the false data are aggregated at the controller. The framework of MAS subjected to FDIA is given in Fig. 1.

Take the Agent i for an example, the false data is injected to its neighbouring agents in network. The controller receives the mixed state value and the false data from the communication networks. Due to the existence of the false data injection, the actual state information of the neighbours from the communication network cannot be utilized anymore. Therefore, the above distributed protocol (1) turns out to be the following one under the attack

$$\begin{aligned} u_i^{(2)}(t) &= \sum_{j=1}^{N} a_{ij}K\Big(x_i(t) - \big(x_j(t) + d_{ij}(t)\big)\Big) \\ &= \sum_{j=1}^{N} a_{ij}K\big(x_i(t) - x_j(t)\big) - Kd_i(t), \end{aligned} \tag{3}$$

where $d_i(t)$ is the received injected false data from its neighbours, denoted as

$$d_i(t) = \sum_{j=1}^{N} a_{ij}d_{ij}(t), \tag{4}$$

and $d_{ij}(t)$ is the injected false data from the communication link e_{ij} $(j \hookrightarrow i)$.

Remark 1: In this paper, the sum of all the false data received by the Agent i from its neighbors is represented as $d_i(t)$. How to separate the false data from different neighbors to make the research more targeted is a problem worth discussing, and we will gradually improve this deficiency in the follow-up work.

In fact, the state information of jth neighbour and the false data information d_{ij} are packaged together. That is, the received state value at the ith agent is $\tilde{x}_j(t) = x_j(t) + d_{ij}(t)$ instead of the original actual state value $x_j(t)$. Under this false data injection scenario, the desired consensus cannot be achieved any more by the distributed protocol (3). In order to solve this problem, it is necessary to design a false data detection algorithm to identify the injected false data and the corresponding defensive strategy is needed to restore the desired consensus control objective.

This paper intends to design a state observer detection algorithm to estimate the injected false data and then proposes a distributed observer-based consensus protocol for the MASs and furthermore analyzes the consensus conditions for the MASs with undirected and directed communication network subjected to two types of false data injection attacks. Some preliminaries are provided as follows:

Assumption 1: The injected false data $d_i(t)$ satisfies the following condition:

$$\dot{d}_i(t) = Wd_i(t),\ i = 1,\ 2,\ \cdots,\ N, \tag{5}$$

where $W \in R^{n\times n}$ is the attacker's dynamic identified by ith agent and the eigenvalues of the matrix W have negative real parts.

Remark 2: In practice, the attackers may send bounded and decaying false data to the system due to resource constraints as Assumption 1. Most of the eigenvalues of

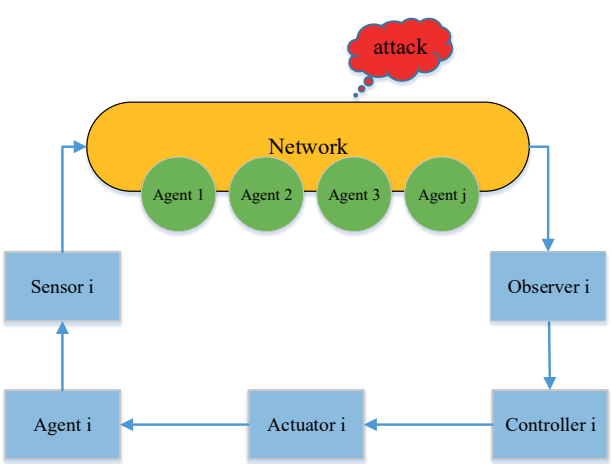

Fig. 1. The framework of MAS subjected to FDIA aggregated at the controller.

the attacker's dynamics matrix W are may selected the same as those of the system dynamics matrix A to make the MASs difficult to detect the occurrence of the attack, so as to destroy the stability of the system.

Assumption 2: The injected false data $d_i(t)$ satisfies the following condition:

$$\dot{d}_i(t) = h_i(t) \text{ and } \lim_{t\to\infty} \|h_i(t)\| = 0, \tag{6}$$

for $i = 1, 2, \cdots, N$.

Remark 3: In the actual control process, the attack dynamics model launched by the attackers is usually unknown, so Assumption 1 has a great limitation. However, the exact attack model is not needed in Assumption 2, which is more practical.

Lemma 1: If the digraph $\mathcal{G}$ contains a directed spanning tree, 0 is a simple eigenvalue of L.

Lemma 2 [50]**:** If the digraph $\mathcal{G}$ contains a directed spanning tree, then with proper permutation, the Laplacian matrix L can be reduced to the Frobenius normal form

$$L = \begin{bmatrix} L_{11} & L_{12} & \cdots & L_{1k} \\ 0 & L_{22} & \cdots & L_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & L_{kk} \end{bmatrix}.$$

where $L_{ii}, i = 1, 2, \cdots, k-1$ are irreducible, each subblock L_{ii} has at least one row with positive row sum, and L_{kk} is irreducible or is a zero matrix of dimension one.

3. MAIN RESULTS

This section provides the theoretical analysis for the resilient control strategies and the consensusable problems of linear MASs under false data injection attacks.

3.1. Linear MASs with undirected topology

To begin with, the resilient consensus control problem is considered for linear MAS with an undirected topology under the bounded and decaying false dada injection attack (Assumption 1). The attack scenario is described by Fig. 2 which can be called the link injection attack. Taking the Agent 3 as the attack target, in order to destroy the consensus performance of the MAS, the adversary tries to inject false data into the communication channel of Agent 3's neighbouring agents. Therefore, the received data information is the mixed actual state values and the the injected false data.

In order to restore the desired consensus performance of MAS under the FDIA, the detection and defence strategies are needed for the reconstruction process. The state observer method is utilized in this paper. The local Luenberger state observer for ith agent is described by

$$\begin{cases} \dot{\hat{x}}_i(t) = A\hat{x}_i(t) + Bu_i(t) + H_1(\hat{y}_i(t) - y_i(t)), \\ \hat{y}_i(t) = C\hat{x}_i(t), \end{cases} \tag{7}$$

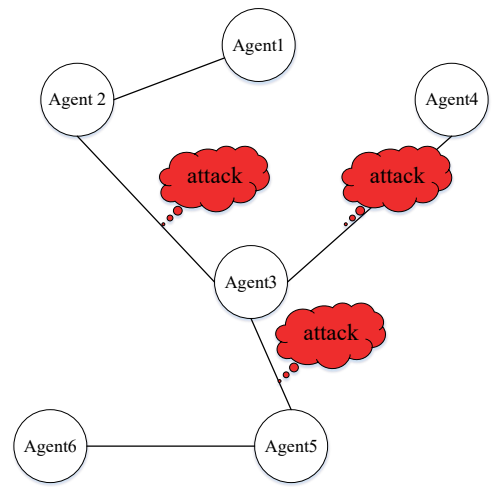

Fig. 2. The FDIA of the undirected MAS.

where $\hat{x}_i(t)$ is the estimation of the state $x_i(t)$, $\hat{y}_i(t)$ is the output of the observer system, and $H_1 \in R^{n\times m}$ is the feedback gain matrix.

The ith agent's false data observer is designed by

$$\dot{\hat{d}}_i(t) = W\hat{d}_i(t) + H_2\big(\hat{y}_i(t) - y_i(t)\big), \tag{8}$$

where $\hat{d}_i(t)$ is the estimation value of the injected attack $d_i(t)$.

Remark 4: Note that the other resilient control approaches in [42-46] can be summarized as passively adjusting the control gain or control frequency to reduce the impacts. This paper designs an attack observer using output signals to counteract the effects of attacks, which can be considered as an active approach.

Based on the observed state value $\hat{x}_i(t)$ for ith agent and the observed false data value $\hat{d}_i(t)$, the following distributed protocol is designed

$$u_i^{(3)}(t) = \sum_{j=1}^{N} a_{ij}K\big(\hat{x}_i(t) - \tilde{x}_j(t)\big) + K\hat{d}_i(t), \tag{9}$$

where $\tilde{x}_j(t)$ is the received state information from jth neighbour, which can be viewed as the combination of $\hat{x}_j(t)$ and $d_{ij}(t)$, i.e., $\tilde{x}_j(t) = \hat{x}_j(t) + d_{ij}(t)$.

Therefore, the distributed protocol (9) can be translated into

$$\begin{aligned} u_i(t) = &\sum_{j=1}^{N} a_{ij}K(\hat{x}_i(t) - (\hat{x}_j(t) + d_{ij}(t))) \\ &+ K\hat{d}_i(t). \end{aligned} \tag{10}$$

By augmenting $x_i(t)$ and $d_i(t)$ into an augmented vector, one can reformulate the system in the following form

$$\dot{\bar{x}}_i(t) = \bar{A}\bar{x}_i(t) + \bar{B}u_i(t), \tag{11}$$

where the augmented vector and matrices are

$$\bar{x}_i(t) = \begin{bmatrix} x_i(t) \\ d_i(t) \end{bmatrix}, \ \bar{A} = \begin{bmatrix} A & -BK \\ \mathbf{0}_{n\times n} & W \end{bmatrix}, \ \bar{B} = \begin{bmatrix} B_{n\times q} \\ \mathbf{0}_{n\times q} \end{bmatrix}.$$

Furthermore, the observer system can be derived as well

$$\dot{\hat{\bar{x}}}_i(t) = \bar{A}\hat{\bar{x}}_i(t) + \bar{B}u_i(t) + H(\hat{y}_i(t) - y_i(t)), \tag{12}$$

where $\hat{\bar{x}}_i$ is the augmented vector and $H \in R^{2n\times m}$ is the augmented feedback gain matrix, given as follows:

$$\hat{\bar{x}}_i(t) = \begin{bmatrix} \hat{x}_i(t) \\ \hat{d}_i(t) \end{bmatrix}, \ H = \begin{bmatrix} H_1 \\ H_2 \end{bmatrix}.$$

The state estimation error and the false data attack estimation error of ith agent are defined as

$$e_{x_i}(t) = \hat{x}_i(t) - x_i(t) \ \text{ and } \ e_{d_i}(t) = \hat{d}_i(t) - d_i(t).$$

Denote $e_i(t) = [e_{x_i}^T, \ e_{d_i}^T]^T \in R^{2n\times 1}$, $e(t) = [e_1^T, \ e_2^T, \ \cdots, \ e_N^T]^T \in R^{2Nn\times 1}$, and $F_1 = [I_n, \ \mathbf{0}_{n\times n}]$, $F_2 = [\mathbf{0}_{n\times n}, \ I_n]$. The following relation between the error vectors are defined

$$e_x^T(t) = [I_N \otimes F_1]e(t) \text{ and } e_d^T(t) = [I_N \otimes F_2]e(t). \tag{13}$$

Furthermore, by augmenting $x(t)$ and $\hat{x}(t)$ with $x(t) = [x_1^T, \ x_2^T, \ \cdots, \ x_N^T]^T$, $\hat{x}(t) = [\hat{x}_1^T, \ \hat{x}_2^T, \ \cdots, \ \hat{x}_N^T]^T$, the coupled network dynamics by (1) and (10) can be rewritten into a compact form as

$$\begin{aligned} \dot{x}(t) = &(I_N \otimes A + L \otimes BK)x(t) + (L \otimes BK)e_x(t) \\ &+ (I_N \otimes BK)e_d(t). \end{aligned} \tag{14}$$

According to (11) and (12), the dynamic of the observer error system $e(t)$ follows

$$\dot{e}(t) = (I_N \otimes \bar{A})e(t) + (I_N \otimes HC)e_x(t). \tag{15}$$

On the other hand, by defining the consensus error vector

$$\varepsilon(t) = (M \otimes I_n)x(t), \tag{16}$$

where $M = I_N - \mathbf{1}\xi_1^T$ and $\xi_1 = \left(\frac{1}{N}, \frac{1}{N}, \cdots, \frac{1}{N}\right)^T$.

From (14) and (16) and the equality $ML = L$, it is easy to derive

$$\begin{aligned} \dot{\varepsilon}(t) = &(I_N \otimes A + L \otimes BK)\varepsilon(t) + (L \otimes BK)e_x(t) \\ &+ (M \otimes BK)e_d(t). \end{aligned} \tag{17}$$

It can be verified that $\varepsilon(t) = 0$ if and only if the consensus is reached.

Theorem 1: As for the linear MASs (1) with the distributed protocol (10), suppose the communication network is undirected and connected and the false data injection attack satisfies Assumption 1, the consensus problem of MASs can be globally asymptotically solved with $K = -B^T P_1$ if there exist positive definite matrices P_1, P_2, positive scalar β such that

$$\begin{cases} P_1 A + A^T P_1 - 2\lambda_2(L)P_1 BB^T P_1 + \beta I_n \le 0, \\ \Omega < 0, \end{cases} \tag{18}$$

where the matrix Ω is given with

$$\begin{aligned} &\Omega = \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ * & \Omega_{22} \end{bmatrix}, \ \Omega_{11} = -\frac{1}{2}\beta I_N \otimes I_n, \\ &\Omega_{12} = -\frac{1}{2}L \otimes (P_1 BB^T P_1 F_1) - \frac{1}{2}M \otimes (P_1 BB^T P_1 F_2), \\ &\Omega_{22} = I_N \otimes (P_2\bar{A} + \bar{A}^T P_2 + P_2 HCF_1 + F_1^T C^T H^T P_2), \end{aligned}$$

where $\lambda_2(L)$ represents the second smallest eigenvalue of Laplacian matrix L.

Proof: Consider the following Lyapunov function

$$V(t) = \frac{1}{2}\varepsilon^T(t)(I_N \otimes P_1)\varepsilon(t) + e^T(t)(I_N \otimes P_2)e(t). \tag{19}$$

Then, the time derivative of the Lyapunov candidate along the trajectories of the error system is

$$\begin{aligned} \dot{V}(t) &= \varepsilon^T(t)(I_N \otimes P_1)\dot{\varepsilon}(t) + 2e^T(t)(I_N \otimes P_2)\dot{e}(t) \\ &= \varepsilon^T(t)(I_N \otimes P_1 A)\varepsilon(t) - \varepsilon^T(t)(L \otimes P_1 BB^T P_1)\varepsilon(t) \\ &\quad - \varepsilon^T(t)(L \otimes P_1 BB^T P_1)e_x(t) \\ &\quad - \varepsilon^T(t)(M \otimes P_1 BB^T P_1)e_d(t) \\ &\quad + 2e^T(t)(I_N \otimes P_2\bar{A})e(t) \\ &\quad + 2e^T(t)(I_N \otimes P_2 HC)e_x(t). \end{aligned}$$

By submitting (13) to the above formulation, one can derive that

$$\begin{aligned} \dot{V}(t) \le &\frac{1}{2}\varepsilon^T(t)\left[I_N \otimes (P_1 A + A^T P_1) - 2L \otimes P_1 BB^T P_1\right]\varepsilon(t) \\ &- \varepsilon^T(t)(L \otimes P_1 BB^T P_1 F_1)e(t) \\ &- \varepsilon^T(t)(M \otimes P_1 BB^T P_1 F_2)e(t) \\ &+ 2e^T(t)(I_N \otimes P_2\bar{A})e(t) \\ &+ 2e^T(t)(I_N \otimes P_2 HCF_1)e(t). \end{aligned}$$

Since the Laplacian matrix L is a real symmetric matrix, there exists an orthogonal matrix U such that

$$U^{-1}LU = U^T LU = J = diag(0, \ \lambda_2, \ \cdots, \ \lambda_{max}),$$

where

$$U = \begin{bmatrix} \frac{1}{\sqrt{N}} & Z_1 \end{bmatrix}, \ U^T = \begin{bmatrix} \frac{1}{\sqrt{N}} \\ Z_2 \end{bmatrix},$$

with $Z_1 \in R^{N\times(N-1)}$ and $Z_2 \in R^{(N-1)\times N}$.

Let $\bar{\varepsilon}(t) = \left[\bar{\varepsilon}_1^T(t), \bar{\varepsilon}_2^T(t), \cdots, \bar{\varepsilon}_N^T(t)\right]^T = (U^T \otimes I_n)\varepsilon(t)$, then one can derive

$$\begin{aligned}&\varepsilon^T(t)\left[I_N \otimes (P_1A + A^TP_1) - 2L \otimes P_1BB^TP_1\right]\varepsilon(t)\\&= \sum_{i=1}^{N} \bar{\varepsilon}_i^T(t)(P_1A + A^TP_1 - 2\lambda_i P_1BB^TP_1)\bar{\varepsilon}_i(t).\end{aligned}$$

Furthermore, one can get

$$\begin{aligned}\dot{V}(t) \le &-\frac{1}{2}\beta\varepsilon^T(t)\varepsilon(t) - \varepsilon^T(t)(L \otimes (P_1BB^TP_1F_1)e(t)\\&- \varepsilon^T(t)(M \otimes P_1BB^TP_1F_2)e(t)\\&+ 2e^T(t)(I_N \otimes P_2\bar{A})e(t)\\&+ 2e^T(t)(I_N \otimes P_2HCF_1)e(t).\end{aligned}$$

Let $\epsilon(t) = \left[\varepsilon^T(t), e^T(t)\right]^T$, it can be verified that

$$\dot{V}(t) \le \epsilon^T(t)\Omega\epsilon(t) < 0, \tag{20}$$

which means the observer error systems (7) and (8) are asymptotically stable. That is, the consensus problem for MASs (1) under the false data injection attack is asymptotically solved. This completes the proof. □

Algorithm 1: Under the assumption that (A, B, C) is controllable and observable, the false data injection attack $d_i(t)$ is bounded and decaying and satisfying $\dot{d}_i(t) = Wd_i(t)$. The attacker's dynamic matrix W, positive definite matrices P_1, P_2, and positive scalar β of the distributed protocol can be designed as

1) The injected false data $d_i(t)$ is stored by ith agent based on the historical data collection, and attacker's dynamic matrix W can be estimated by least square estimation.
2) Since (A, C) is observable, H_1 can be chosen to ensure the eigenvalues of $A + H_1C$ have negative real parts.
3) Solve the following linear matrix inequality

$$\begin{bmatrix} P_1A + A^TP_1 + \beta I_n & P_1B \\ B^TP_1 & -I_n \end{bmatrix} < 0, \tag{21}$$

to derive the positive scalar β and positive definite matrix P_1 such that $P_1A + A^TP_1 - 2\lambda_2(L)P_1BB^TP_1 + \beta I_n \le 0$ by schur complement lemma.
4) Finally, solve the linear matrix inequality $\Omega < 0$ in (18), and the positive matrix P_2 is obtained.

Previously, the false data injection is assumed to be a bounded and decaying one and its dynamic follows a linear system. In the actual control system, the false data model is usually unknown, so it is difficult to estimate the attacker's dynamics matrix. In the following, we consider another kind of false data injection model for directed communication networks.

3.2. Linear MASs with directed topology

This subsection considers the false data injection attack with a decaying rate (Assumption 2) for directed linear MASs. Different from the link injection attack for undirected MAS, the attack for a directed topology can be illustrated as the node injection attack. The attack scenario is described by the following Fig. 3. For example, if the attacker takes the Agent 3 as an attack target, then the injected false data will be delivered to its neighbouring agents and the the emanative data information is the mixed actual state values and the the injected false data.

Similar to the previous subsection, the system (1) under the injected attack with decaying rate can be reformulated in the following form

$$\dot{\bar{x}}_i(t) = \bar{A}\bar{x}_i(t) + \bar{B}u_i(t) + Qh_i(t), \tag{22}$$

where $\bar{x}_i$ and $\bar{B}$ are the same with the previous ones and the matrices

$$\bar{A} = \begin{bmatrix} A & -BK \\ \mathbf{0}_{n\times n} & \mathbf{0}_{n\times n} \end{bmatrix}, \; Q = [\mathbf{0}_{n\times n}, I_n]^T.$$

In this case, the false data observer is designed directly as

$$\dot{\hat{d}}_i(t) = H_2(\hat{y}_i(t) - y_i(t)), \tag{23}$$

where $H_2 \in R^{n\times m}$ is the feedback gain matrix.

Furthermore, the observer for the augmented system is designed for each agent

$$\dot{\hat{\bar{x}}}_i(t) = \bar{A}\hat{\bar{x}}_i(t) + \bar{B}u_i(t) + H(\hat{y}_i(t) - y_i(t)), \tag{24}$$

where $H = [H_1^T, H_2^T]^T$ is the feedback gain matrix.

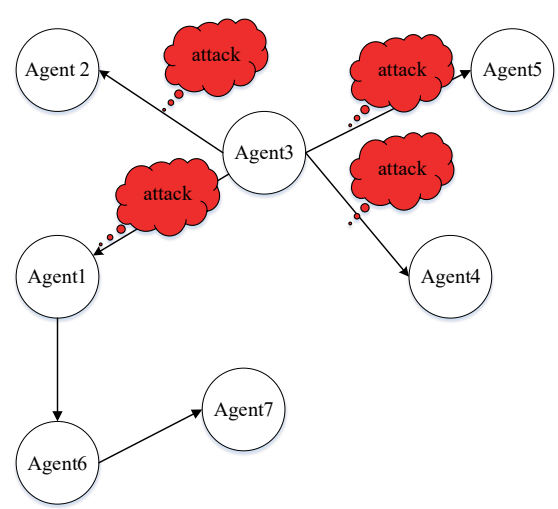

Fig. 3. The node injection attack for linear MAS with directed topology.

According to (22) and (24), the dynamic of the error $e(t)$ follows

$$\dot{e}(t) = (I_N \otimes \bar{A})e(t) + (I_N \otimes HC)e_x(t) - (I_N \otimes Q)h(t), \tag{25}$$

where $h(t) = \left[h_1^T, h_2^T, \cdots, h_N^T\right]^T$.

Similarly, by letting the consensus error variable $\varepsilon(t) = (M \otimes I_n)x(t)$, where $M = I_N - \mathbf{1}\xi_2^T$, where ξ_2 is the left eigenvector of the Laplacian matrix L with the corresponding eigenvalue 0, and $\xi_2^T\mathbf{1} = 1$.

Because of the vectors ξ_2 and $\mathbf{1}$ are the left eigenvector and right eigenvector of the Laplacian matrix L respectively, one can derive that

$$\begin{aligned}\dot{\varepsilon}(t) &= (I_N \otimes A + L \otimes BK)\varepsilon(t) + (L \otimes BK)e_x(t) \\ &\quad + (M \otimes BK)e_d(t).\end{aligned} \tag{26}$$

Motivated by [49], a necessary and sufficient condition has been provided for solving the consensus problem.

Theorem 2: Under the assumption that the communication topology is directed and contains a directed spanning tree, the consensus of the linear MASs with decaying rate injected attack can be reached if and only if all matrices $A + \lambda_i BK, \bar{A} + HCF_1$ are Hurwitz ones, where λ_i, $i = 2, \cdots, N$ are the nonzero eigenvalues of the Laplacian matrix L.

Proof: By combining the observer error system (25) and the consensus error system (26), the compact form error system turns out to be

$$\begin{bmatrix}\dot{\varepsilon}(t) \\ \dot{e}(t)\end{bmatrix} = \Psi \begin{bmatrix}\varepsilon(t) \\ e(t)\end{bmatrix} - \begin{bmatrix}\mathbf{0} \\ I_N \otimes Q\end{bmatrix} h(t), \tag{27}$$

$$\Psi = \begin{bmatrix} I_N \otimes A + L \otimes BK & L \otimes BKF_1 + M \otimes BKF_2 \\ \mathbf{0} & I_N \otimes (\bar{A} + HCF_1)\end{bmatrix}.$$

According to Lemma 2, there exists an invertible matrix P such that the Laplacian matrix L can be converted to Jordan normalized form as

$$P^{-1}LP = J = \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & \triangle \end{bmatrix},$$

where the diagonal elements of $\triangle$ are the nonzero eigenvalues of L.

Let $\gamma(t) = (P^{-1} \otimes I_n)\varepsilon(t)$, and $\eta(t) = (P^{-1} \otimes I_{2n})e(t)$, then (27) can be represented as

$$\begin{bmatrix}\dot{\gamma}(t) \\ \dot{\eta}(t)\end{bmatrix} = \Psi' \begin{bmatrix}\gamma(t) \\ \eta(t)\end{bmatrix} - \begin{bmatrix}\mathbf{0} \\ P^{-1} \otimes Q\end{bmatrix} h(t), \tag{28}$$

where Ψ' is given with

$$\Psi' = \begin{bmatrix} I_N \otimes A + J \otimes BK & J \otimes BKF_1 + M_1 \otimes BKF_2 \\ \mathbf{0} & I_N \otimes (\bar{A} + HCF_1)\end{bmatrix},$$

$$M_1 = P^{-1}MP = \begin{bmatrix}\tau \\ M_2\end{bmatrix}, \tau \in R^{1\times N}, M_2 \in R^{(N-1)\times N}.$$

By letting $\gamma(t) = [\gamma_1(t)^T, \bar{\gamma}(t)^T]^T$, one can get $\gamma_1(t) = (\xi_2^T \otimes I_n)\varepsilon(t) \equiv \mathbf{0}$. Thus, (28) can be simplified to the reduced order one

$$\begin{bmatrix}\dot{\bar{\gamma}}(t) \\ \dot{\eta}(t)\end{bmatrix} = \Psi_1' \begin{bmatrix}\bar{\gamma}(t) \\ \eta(t)\end{bmatrix} - \begin{bmatrix}\mathbf{0}_{(N-1)n\times Nn} \\ P^{-1} \otimes Q\end{bmatrix} h(t), \tag{29}$$

where the coefficient matrix Ψ_1' is

$$\Psi_1' = \begin{bmatrix} I_{N-1} \otimes A + \triangle \otimes BK & \Psi_{12}' \\ \mathbf{0} & I_N \otimes (\bar{A} + HCF_1)\end{bmatrix},$$

where $\Psi_{12}' = \triangle_1 \otimes BKF_1 + M_2 \otimes BKF_2$ and $\triangle_1 = [\mathbf{0}, \triangle]$.

It should be noted that the elements of the state matrix Ψ_1' are block upper triangular. Therefore, when the matrices $A + \lambda_i BK, \bar{A} + HCF_1$ are Hurwitz, Ψ_1' is Hurwitz one as well. By Assumption 1, $\lim_{t\to\infty} \|h_i(t)\| = 0$, the $\gamma(t)$, $\eta(t)$ will converge to zero as well. Hence the proof is completed. □

Remark 5: According to Lemma 1, since the transformed Jordan normalized form J contains a zero eigenvalue, the Hurwitz criterion cannot be used. Therefore, it is necessary to reduce the order through transformation to eliminate the influence of the zero eigenvalue.

Algorithm 2: As for the linear MASs with (A, B, C) controllable and observable under the injected false data attack with a decrying rate (Assumption 2), if $\mathcal{G}$ contains a directed spanning tree, then the gain matrix K, the positive definite matrix P, and the feedback gain matrix H can be designed as

1) Choose K to satisfy $A + \lambda_i(L)BK$ is Hurwitz.
2) Solve the matrix inequality $(\bar{A} + HCF_1)^T P + P(\bar{A} + HCF_1) < 0$, one can derive $P > 0$. Then, H can be chosen as $H = -P^{-1}F_1^T C^T$.

4. ILLUSTRATIVE EXAMPLES

This section presents two illustrative examples to demonstrate the effectiveness of the proposed distributed protocol against the false data injection attack.

Example 1: Consider a linear MAS with ten agents and the undirected communication topology is shown in Fig. 4.

The system parameter matrices are given as

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -3 \end{bmatrix}, B = \begin{bmatrix}0 \\ 2\end{bmatrix}, C = \begin{bmatrix}-1 & 3\end{bmatrix}.$$

It is easy to check that (A, B, C) is controllable and observable. Let the initial states of the systems (1) and (7) are chosen randomly. The initial states of ten false

data are [-1.5,1.5,1,2.4,3,2.1,-3,4.4,2.8,-2;2,3,-3,2,1,3,3,-4.5,3.3,3]. According to Algorithm 1, one can derive attacker's dynamic matrix

$$W = \begin{bmatrix} -0.7027 & -0.2344 \\ 0.3909 & -1.6048 \end{bmatrix},$$

by the least squares method.

By Algorithm 1, it is easy to derive $H = [1, -3, 3.1, -1]^T$.

By setting the parameter $\beta = 0.1$ and solving the LMI (21) to get the positive denefinite matrix

$$P_1 = \begin{bmatrix} 0.8156 & 0.2384 \\ 0.2384 & 0.3459 \end{bmatrix}.$$

By solving the $\Omega < 0$ to derive the positive definite matrix

$$P_2 = \begin{bmatrix} 999.3620 & 160.4749 & -167.2376 & 62.9466 \\ 160.4749 & 177.1028 & 133.7376 & 48.0490 \\ -167.2376 & 133.7376 & 272.5302 & 89.3847 \\ 62.9466 & 48.0490 & 89.3847 & 236.7725 \end{bmatrix}.$$

The state trajectories and the observed state trajectories of the systems (1) and (7) are shown in Fig. 5, respectively. The false data trajectories and the observed false data trajectories of the systems (5) and (8) are shown in Fig. 6, respectively.

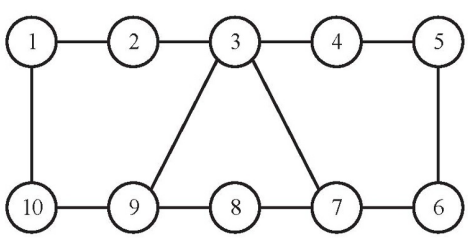

Fig. 4. The network topology of the multi-agent system in undirected graph.

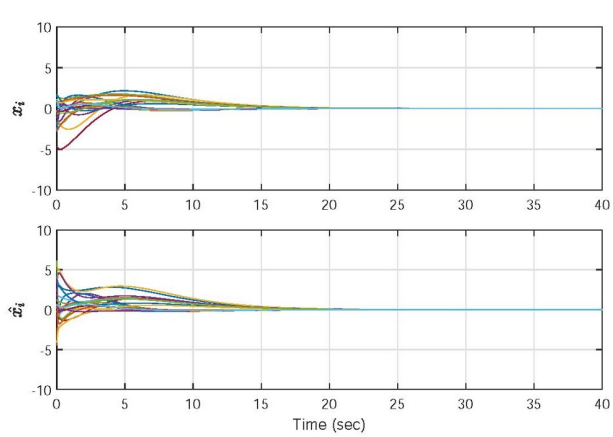

Fig. 5. State trajectory $x(t)$ and observed state $\hat{x}(t)$ by Algorithm 1.

The state observation error as well as the false data observation error are given respectively in Fig. 7 under $\beta = 0.5$.

From Figs. 5 and 7, one can see that the consensus of MAS can be reached among the agents by the proposed consensus algorithm.

Example 2: Consider a linear digraph MAS consisting of ten agents. The topology is shown in Fig. 8.

The parameter matrices are given

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}, C = \begin{bmatrix} -1 & 3 \\ 1 & 0 \end{bmatrix}.$$

By Algorithm 2, one can choose

$$K = \begin{bmatrix} -1 & -0.5 \\ -1 & 0 \end{bmatrix},$$

such that the matrix $A + \lambda_i(L)BK$ is Hurwitz. The false data of each agents are

$$d_1 = \begin{bmatrix} e^{-5t}+2 \\ e^{-t}+1 \end{bmatrix}, d_2 = \begin{bmatrix} e^{-3t}+1 \\ e^{-3t}+2 \end{bmatrix}, d_3 = \begin{bmatrix} e^{-4t}-2 \\ e^{-2t}+1.5 \end{bmatrix},$$

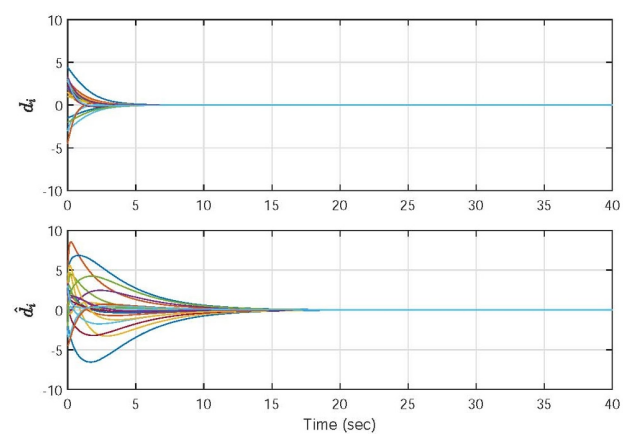

Fig. 6. False data $d(t)$ and observed false data $\hat{d}(t)$ by Algorithm 1.

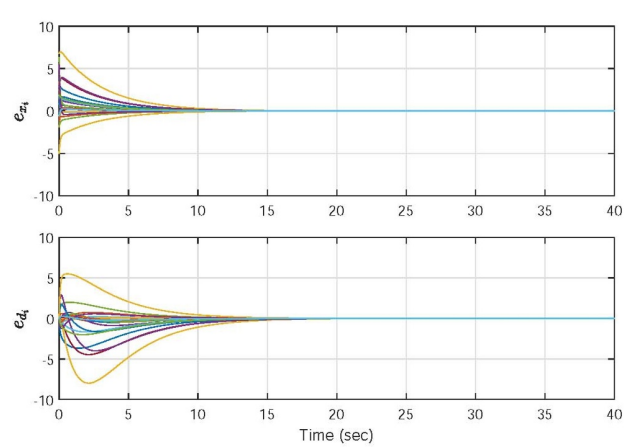

Fig. 7. Estimation error $e(t)$ and false data estimation error $e_d(t)$ by Algorithm 1.

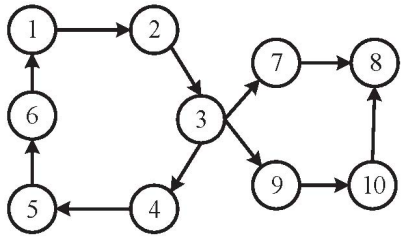

Fig. 8. The network topology of the MAS with a directed graph.

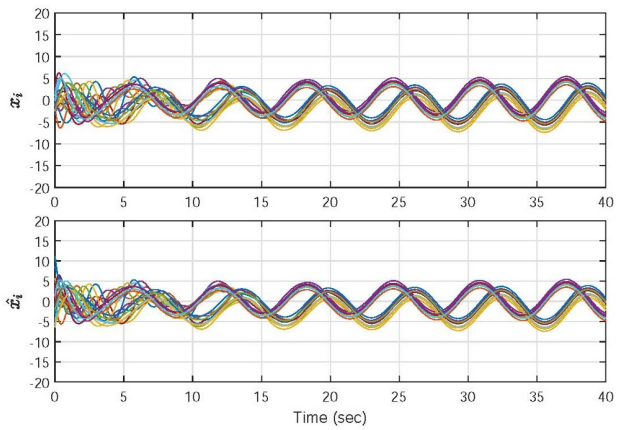

Fig. 9. State trajectory $x(t)$ and observed state $\hat{x}(t)$ by Algorithm 2.

$$d_4 = \begin{bmatrix} e^{-2.5t}+1.5 \\ e^{-5t}-1 \end{bmatrix},\ d_5 = \begin{bmatrix} e^{-t}+3 \\ e^{-3t}+1 \end{bmatrix},\ d_6 = \begin{bmatrix} e^{-5t}-3 \\ e^{-4t}-2 \end{bmatrix},$$
$$d_7 = \begin{bmatrix} e^{-1.5t}+2 \\ e^{-2t}+3 \end{bmatrix},\ d_8 = \begin{bmatrix} e^{-3t}+2 \\ e^{-3t}+4 \end{bmatrix},\ d_9 = \begin{bmatrix} e^{-4t}+1 \\ e^{-5t}-2 \end{bmatrix},$$
$$d_{10} = \begin{bmatrix} e^{-2t}+1.2 \\ e^{-2t}+1 \end{bmatrix}.$$

The initial states of ten false data are $[3,\ 2,\ -1,\ 2.5,\ 4,\ -2,\ 3,\ 3,\ 2,\ 2.2,\ 2,\ 3,\ 2.5,\ 0,\ 2,\ -1,\ 4,\ 5,\ -1,\ 2]^T$. Let the initial state of the systems (1) and (7) are chosen randomly. According to Algorithm 2, one can derive the positive definite matrix

$$P = \begin{bmatrix} 1.2873 & -0.4816 & -1.8253 & 0.2102 \\ -0.4816 & 2.8068 & -0.0678 & -0.8381 \\ -1.8253 & -0.0678 & 7.2508 & -0.0404 \\ 0.2102 & -0.8381 & -0.0404 & 5.3061 \end{bmatrix},$$

and the feedback gain matrix

$$H = \begin{bmatrix} 0.6443 & -1.3539 \\ -1.0095 & -0.2365 \\ 0.1517 & -0.3430 \\ -0.1838 & 0.0137 \end{bmatrix}.$$

The state trajectories and the observed state trajectories of the systems (1) and (7) are shown in Fig. 9, respectively.

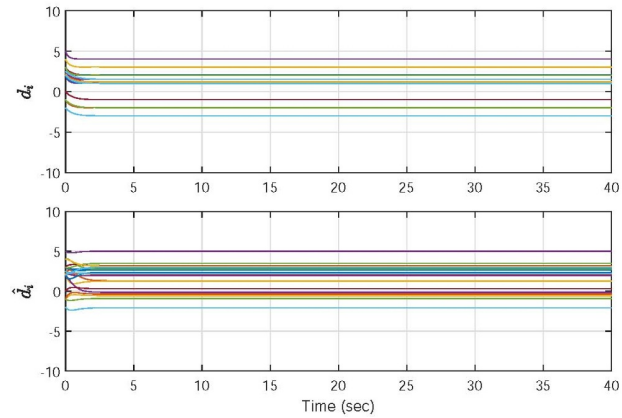

Fig. 10. False data $d(t)$ and observed false data $\hat{d}(t)$ by Algorithm 2.

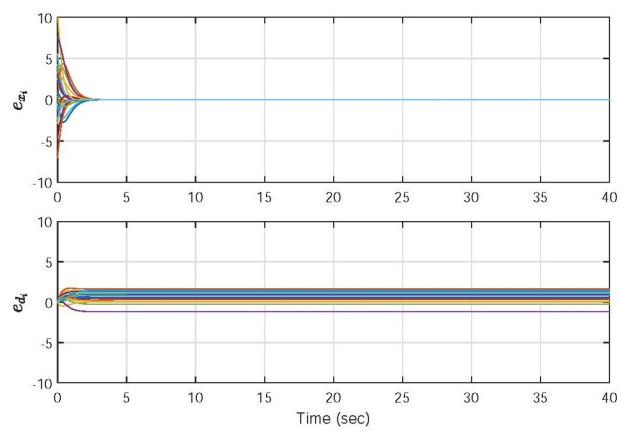

Fig. 11. Estimation error $e(t)$ and false data estimation error $e_d(t)$ by Algorithm 2.

The false data trajectories and the observed false data trajectories of the systems (6) and (23) are shown in Fig. 10, respectively.

The state estimation error and false data observation error are given in Fig. 11. As can be seen from the error trajectories, the state estimation system is asymptotically stable. However, the false data estimation error system is bounded but not asymptotically stable. In this case, the asymptotical consensus can not be achieved while the resilient consensus can be achieved with a small error bounds.

5. CONCLUSION

This paper studied the resilient consensus control problem of linear MAS with the FDIA form the communication networks. An observer-based distributed algorithm was designed to offset the negative effect of the cyber attacks. When the false data model is linear and decaying, a sufficient condition is obtained to ensure the consensus

performance for undirected MAS. Furthermore, we considered the general false data model with a decaying rate and provided a sufficient and necessary condition for the resilient consensus of directed MAS. However, this paper is only limited to attenuating attacks in linear MAS. In fact, most systems are nonlinear and the attacks launched by the attackers are stochastic. The future work will focus on solving the consensus problems of nonlinear MAS with the more actual FDIA models when the attack is on the partial of the critical nodes or links of the communication networks.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with respect to the research, authorship, and/or publication of this paper.

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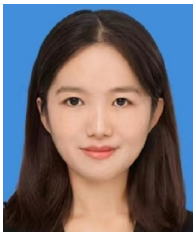

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